AIMS Cameroon MSc Data Science Group Assignment

Course: Deep Learning Algorithms and Methods
Instructor: Prof. Dr. Olushina Olawale Awe

1. Assignment Overview

This group assignment is designed to assess students' understanding of modern **deep learning models**, combining **theoretical foundations**, **mathematical formulation**, applications, and **practical implementation** using **real-world datasets**. Students will work in groups to study, implement, and critically evaluate a specific deep learning architecture. Each group is expected to demonstrate both *conceptual mastery* and *applied competence* through presentations, written reports, and reproducible codes.

2. Group Structure and Assigned Topics

The class is divided into **seven (7) groups**. Each group is randomly assigned one topic as follows:

Group	Topic
Group 1	Transformers
Group 2	Autoencoders
Group 3	Convolutional Neural Networks (CNN)
Group 4	Feedforward Neural Networks (FFNN) / Multilayer Perceptrons (MLP)
Group 5	Reinforcement Learning (RL)
Group 6	Long Short-Term Memory Networks (LSTM)
Group 7	Recurrent Neural Networks (RNN) and GRU

3. Learning Objectives

Upon successful completion of this assignment and course, students should be able to:

- Explain deep learning architectures conceptually and mathematically
- Implement models using real-world datasets
- Understand training dynamics, loss functions, and optimization methods
- Evaluate model performance using appropriate metrics
- Communicate results clearly through presentations and technical reports
- Critically reflect on model limitations and ethical considerations

4. Data Requirement

Each group must use **real, externally sourced data**. Acceptable sources include:

- Public datasets (e.g., Kaggle, UCI Machine Learning Repository, OpenML)
- Domain-specific datasets (health, text, images, climate, economics, education)

For the dataset used, students must clearly describe:

- Data source and context
- Input and output variables
- Preprocessing steps
- Ethical or bias considerations (where applicable)

5. Required Deliverables

Each group must submit the following **three components**.

A. Group Presentation (10–15 minutes)

The presentation must include:

1. Introduction and problem motivation
2. Conceptual explanation of the model
3. Mathematical formulation of the model
4. Dataset description and preprocessing
5. Model architecture and training procedure
6. Application, results, evaluation metrics, and visualizations
7. Critical discussion of results
8. Conclusion and key lessons learned

B. Written Report (10–15 pages)

The report must follow an academic structure:

1. Abstract
2. Introduction
3. Background / Related Work
4. Model Theory and Mathematics
5. Dataset Description and Preprocessing

6. Application and Experimental Setup
7. Results and Analysis
8. Discussion and Limitations
9. Ethical Considerations (if applicable)
10. Conclusion and Future Work
11. References

C. Code Submission

Each group must submit reproducible code, including:

- Clean and well-commented Python codes
- Jupyter notebooks or scripts
- A `README` file explaining how to run the code

6. Model-Specific Expectations

Each group should emphasize aspects unique to their assigned model as distributed below:

- **FFNN / MLP:** feature learning, depth vs. width, non-linearity
- **RNN:** sequence modeling, temporal dependencies
- **CNN:** convolution, filters, pooling, spatial structure
- **LSTM:** gates, memory control, long-term dependencies
- **Autoencoders:** latent representations, reconstruction loss
- **Transformers:** attention mechanisms, self-attention
- **Reinforcement Learning:** agent, environment, reward, policy learning

7. Assessment Criteria

Criterion	Weight
Conceptual understanding	20%
Mathematical clarity	15%
Data selection and preprocessing	15%
Model implementation	20%
Results and evaluation	15%
Presentation quality and teamwork	10%
Code quality and reproducibility	5%

8. Academic Integrity

All sources of data, code, and literature must be properly cited. Plagiarism or uncredited reuse of materials or OpenAI will attract penalties in accordance with university regulations.

9. Submission Format and Deadline

- Presentation slides: PDF
- Written report: PDF
- Code: ZIP file or Git repository link
- Submission deadline: **Saturday 6:30 AM**

10. Final Remark

This assignment goes beyond running algorithms. Students are expected to **understand, explain, justify, and critically evaluate** deep learning models as scientific tools for solving real-world problems.

"A good model is not only one that performs well, but one that can be explained and trusted."
Best wishes!